

3-DAY CHECKLIST FOR 65+ SCORE

DAY 1 — CORE FOUNDATION

◆ BLOCK 1: Basics + Bias-Variance

- Tom Mitchell Definition (T, P, E)
- Supervised vs Unsupervised vs RL
- Classification vs Regression
- Overfitting vs Underfitting
- Bias vs Variance
- Effect of K in KNN
- Solve all related mock questions

◆ BLOCK 2: Clustering (VERY IMPORTANT)

- K-Means steps
- K-Means objective (WCSS)
- Why multiple runs (local minima)
- Elbow Method
- Hierarchical Clustering:
 - Single Link
 - Complete Link
- DBSCAN:
 - Core point
 - Border point
 - Outlier
 - ϵ and MinPts
- Solve all clustering questions

◆ BLOCK 3: Decision Trees

- Entropy
- Gini Index
- Information Gain
- Overfitting in trees
- Early stopping vs Pruning

- Solve tree-based questions

◆ **END REVISION**

- Revise all formulas + definitions
- Review wrong MCQs

■ **DAY 2 — MODELS + SCORING BOOST**

◆ **BLOCK 1: Regression**

- Linear Regression (concept)
- Ridge Regression (L2)
- Lasso Regression (L1)
- Multicollinearity

◆ **BLOCK 2: Logistic + LDA/QDA + Bayes**

- Logistic Regression (sigmoid)
- Why linear decision boundary
- LDA vs QDA
- Bayes classifier rule
- Solve MCQs

◆ **BLOCK 3: SVM + Ensembles**

- SVM margin concept
- C parameter
- Hinge loss (basic idea)
- Bagging
- Random Forest (feature randomness)
- Boosting (basic idea)
- Solve MCQs

◆ **BLOCK 4: Evaluation Metrics**

- Accuracy
- Precision
- Recall
- Imbalanced data issue
- Solve MCQs

◆ END REVISION

- Revise:
 - Lasso vs Ridge
 - Logistic regression
 - SVM
 - Trees

■ DAY 3 — FINAL BOOST + PRACTICE

◆ BLOCK 1: Neural + KNN

- Perceptron limitation (XOR)
- Non-linear activation
- Weight initialization issue
- k-NN quick revision

◆ BLOCK 2: Dimensionality + Advanced Clustering

- PCR vs PLS
- BIRCH (drift problem)
- CURE (shrinking step)

◆ BLOCK 3: Mock Practice

- Solve 1 full mock paper
- Solve another (if time)
- Analyze mistakes

◆ BLOCK 4: Weak Area Fix

- Revise weak topics only

🔥 FINAL REVISION (LAST HOURS)

- Bias-Variance
- K-Means + DBSCAN
- Decision Trees
- Lasso vs Ridge
- Logistic Regression
- Random Forest
- SVM (C parameter)